

1

Consider the following attack:

The attacker obtains the public key (k_1, k_2) from the KeyGen of Σ

The attacker chooses two distinct messages m_0 and m_1 from the message space M .

The attacker requests the encryption of the two messages from the encryption oracle of Σ , and receives the ciphertexts (c_1, c_2) and (c'_1, c'_2) , respectively.

The attacker submits the two ciphertexts to the decryption oracle of Σ

The decryption oracle returns either m_0 or m_1 , depending on whether $c_1 = c'_1$ or $c_1 \neq c'_1$, respectively.

The attacker outputs the chosen message that was decrypted by the oracle.

This attack works because the decryption oracle of Σ' uses the decryption oracles of Σ_1 and Σ_2 to decrypt the ciphertexts. Since Σ_1 and Σ_2 have the same message space M , the attacker can choose two messages m_0 and m_1 such that $\Sigma_1.\text{Dec}(k_1, c_1) = m_0$ and $\Sigma_2.\text{Dec}(k_2, c_2) = m_1$, or $\Sigma_1.\text{Dec}(k_1, c_1) = m_1$ and $\Sigma_2.\text{Dec}(k_2, c_2) = m_0$. Then, the attacker can construct two ciphertexts that differ only in the first component, and submit them to the decryption oracle of Σ'

The decryption oracle will either decrypt the first component of the ciphertexts to m_0 or m_1 , depending on which ciphertext was submitted, and decrypt the second component of the ciphertexts to the other message. Therefore, the decryption oracle will always return one of the chosen messages, and the attacker can determine which one by checking which ciphertext was returned. This attack breaks the CCA security of Σ' .

2

$\text{Dec}(k, (r, x))$:

x' is computed using $F(k, m \oplus r) \oplus r$.

then outputs plain text $m = x' \oplus r$

The scheme does not have CCA security because it is not secure when an attacker is given the encryption of two plaintexts m_1 and m_2 . The attacker can construct a new ciphertext $(r, F(k, m_1 \oplus r) \oplus r \oplus F(k, m_2 \oplus r) \oplus r)$ and can decrypt it to obtain $m_1 \oplus m_2$